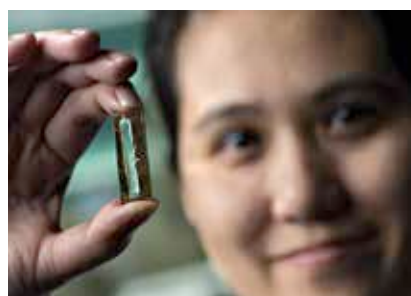


08 BETTERIES

We've already spoken about new energy, but 2017 isn't only going to be about solar panels and wind-mills... it's also going to be about batteries. We know... you're fed up of being promised better batteries (betteries?) for a decade or more, and only ending up with cutting-edge batteries that explode – sorry Samsung, as much as you want to forget 2016, it's going to take more than one year for that debacle to be forgotten.

The reason why nanowire-batteries (which have been promised as the alternative to existing batteries for ages) have not taken off is because the nanowires that store the charge get brittle with every charge and break.



SOURCE: WIKIPEDIA
PHOTO CREDIT: STEVE ZYLJUS/UCL

"Nanowires" can store 10 times more energy than lithium-ion batteries.

This reduces the efficiency of the battery, which in turn means you end up with a battery that loses charge faster and faster beyond a point (usually after 500 cycles, or so). The advantage of nanowires is that they are able to store 10 times the amount of charge as lithium-ion batteries, which means either batteries that last much longer, or much, much smaller and lighter batteries for the same charge.

In a lab in the University of California, Irvine, Mya Le Thai, a PhD candidate, accidentally discovered a way to prevent the nanowires from breaking. Although she worked with gold nanowires, she is confident that this method will work with nickel wires as well. In preliminary testing (which is still underway), she's already charged and discharged the nano-battery over 200,000 times. How does a battery that lasts charging and recharging for 400 years or more sound? It's not

just smartphones that benefit either, it's electric cars, which would be able to increase their driving range on a single charge by 10 times, which is a massive shot in the arm for them.

Another development being talked about (link to video below) is a University of Bristol video and press release that aims to solve a small part of the world's nuclear waste problem (specifically graphite waste from reactors), by making carbon-14 waste into a radioactive diamond, encasing that inside a regular diamond shell to seal in the radioactivity, and then using that as a battery. Sounds like sci-fi or something from a game (alien crystals used for power?), no this is actually very valid science. However, you'd need some humongous radioactive diamonds to run anything substantial. The only awesome thing for really low powered circuits (such as those on satellites or unmanned space vehicles) is that the batteries will run for thousands of years, given that the half-life of carbon-14 is 5,730 years!

FAST BYTES

<http://dgit.in/HAMRv>
<http://dgit.in/3DxPoint>
<http://dgit.in/HAMRfrbs>

09 STORAGE

Though SSDs have matured over the years, they come with one inherent issue, the NAND memory suffers from a limited set of read/write cycles. Hard drives can be re-written millions of times over, and we know of big bulky hard drives from the 70s that are still operational today! With SSDs, however, you have a metric called MTBF (Mean Time Before Failure) which ranges somewhere around a few million hours of operating time. TBW or Total Bytes Written is a more accurate metric to figure out when your SSD will die. Right now, SSDs can only survive 5-6 years of moderate-high volume usage which is way short compared to HDDs.

Enter 3D Xpoint (three-dee-cross-point), which is a non-volatile storage technology that boasts of a lot more read/write cycles than SSDs. With information still limited about this technology, we can't add much more than that, but we'll see consumer grade

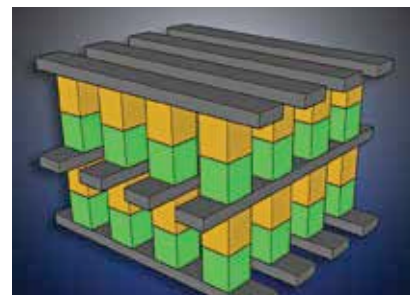
3D Xpoint storage medium in 2017. A supposed evolution of 3D NAND technology, 3D Xpoint is claimed to be faster, durable and power efficient. By how much? Supposedly 4x better

write speeds, 3x better read speeds, 3x write endurance, 10x lower latency and consuming just 30% power. Of course they will probably be 10+3+4+3=20 times more expensive! (We're

kidding!). Historically newer storage technology have only been introduced for the enterprise segment first and consumer SKUs arrive a good year or two later when the costs reduce. So you can drool while we get to benchmark these beauties when they're released.

2017 is also when HDDs will finally lose their commodity status. The density race is boring now, and improvements in hard drive technology are merely cosmetic. Even brands have realised this and have resorted to marketing tactics such as rebranding and segmented targeting.

This is where HAMR (heat assisted magnetic recording) comes to the



SOURCE: WIKIPEDIA

3D Xpoint layer diagram

rescue. The magnetic particles on platters have a property called coercivity, which is a measure of their resistance to an external magnetic field. As you make these particles finer, it becomes more difficult to switch their alignment (from 0 to 1 or vice versa). Thus, you need to reduce their coercivity. Heating is one way to do this, and HAMR uses a fine laser beam to heat the area being written to. Hard drives are going to see a large increase in capacities thanks to HAMR, there'll now be a phase where HDDs become as competitive as SSDs before going back to being commodities in the distant future.

BATTERY BOOM?

<http://dgit.in/ChemBtry>
<http://dgit.in/NuclrBtry>